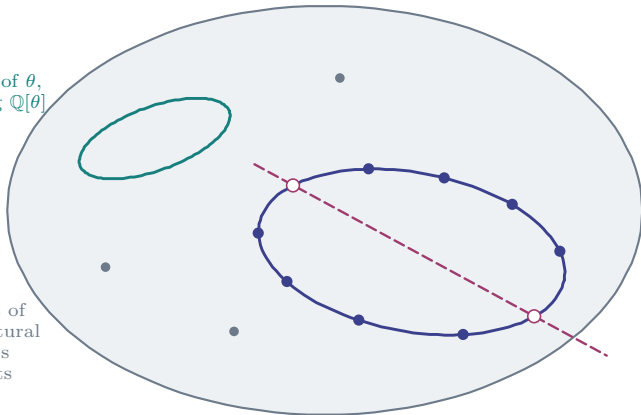


minimal support at $n = 4$, X very general

C_θ : the powers of θ ,
a conic spanning $\mathbb{Q}[\theta]$

Q : the quadric of
Lagrangians; natural
objects are its
rational points



ℓ_W : the Weil line, meeting Q in
the conjugate points $L_\pm$

a plane through ℓ_W cuts a smooth
conic: eight graph subtori B_i on it,
 $\sum m_i[B_i] = 14 W_2$ at $d = 1$

a nonzero Weil part needs at least eight rational points off C_θ , and eight suffice only on such a conic